Does Functional Similarity Induced by Distillation Preserve Internal Representations and Explanatory Mechanisms in Autoregressive Models?

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***Abstract*—Knowledge distillation is often evaluated in terms of functional similarity between teacher and student models, typically measured through predictive performance. However, it remains unclear whether this functional similarity also implies the preservation of internal representations and explanatory mechanisms, or whether it is simply a consequence of training a smaller model on the same task. This work investigates this question in autoregressive Transformer-based models. A TinyLlama teacher is distilled by depth into a family of 11-layer students spanning a full non-distilled baseline, a weakly-distilled variant, a standard-KD variant, and a class-restricted-KD variant, isolating the effect of distillation strength from that of depth reduction alone. Internal representations are compared through layer-wise and global attention-map metrics, spectral analysis, per-head similarity, and linear CKA on hidden states, while explanatory mechanisms are evaluated through LIME, SHAP, and Integrated Gradients (using a zero-vector baseline), moving beyond qualitative examples to systematic, aggregate agreement and faithfulness metrics computed over held-out samples. The study is conducted first as a pilot on binary IMDb sentiment classification and then, as the main experiment, on the four-class, non-binary AG News topic classification task. The results show that distillation does not transfer the teacher's explanatory mechanisms, as all students assign non-trivial importance to tokens lacking task-relevant content, regardless of classification performance. Regarding internal representations, a partial transfer of structure is observed in the first attention layer, with Frobenius distances below 2% of the theoretical maximum bound, whereas intermediate and deeper layers diverge substantially; critically, the non-distilled baseline is often as similar to the teacher's attention and hidden-state representations as the distilled students are, indicating that much of this partial similarity is attributable to shared architecture and task rather than to distillation itself. Distillation's most robust, reproducible effect across both datasets is instead to pull the student's output distribution toward the teacher's, largely independent of whether it improves classification accuracy. Taken together, these results indicate that classification performance, distributional similarity, explanatory alignment, and representational alignment are largely dissociable properties of a distilled model.**

I. Introduction

Explainable Artificial Intelligence (XAI) encompasses a set of methods and processes aimed at interpreting and explaining the behavior of complex mathematical models. In many application domains, it is essential for deployed systems to be interpretable, enabling the justification of autonomous decisions and fostering user trust.

In recent years, autoregressive models based on the Transformer architecture have revolutionized the field of Natural Language Processing (NLP). Large-scale generative models such as GPT, Claude, and Mistral have demonstrated remarkable capabilities in natural language generation and understanding tasks. Despite their strong performance, these models incur substantial computational costs during both training and inference due to their large number of parameters.

To reduce these costs, several model compression techniques have been proposed, among which knowledge distillation (KD) is one of the most widely adopted. In this approach, a student model is trained to approximate the probability distributions produced by a teacher model, with the goal of reproducing its behavior using a significantly smaller architecture [6].

Although distillation aims to preserve the functional behavior of the original model, it remains unclear to what extent the teacher's internal representations and explanatory mechanisms emerge or are reconstructed in the student as a consequence of enforcing similarity in the output distributions. In particular, it is natural to ask whether properties associated with attention mechanisms and prediction explanations are preserved throughout the compression process.

The objective of this work is to investigate the relationship between functional similarity and representational similarity in Transformer models compressed through knowledge distillation, and, critically, to disentangle how much of that relationship is due to distillation itself as opposed to simply training a smaller model on the same task. To this end, we consider a generative classification problem [3] in which the target classes correspond to single tokens in the teacher model's vocabulary. Within this setting, we analyze and compare both internal representations and explanatory mechanisms of the teacher and a family of student models spanning a non-distilled baseline, a weakly-distilled variant, and two distilled variants of increasing fidelity. The motivation for this study stems from the work of [7], which shows that multiple attention configurations can induce similar predictions, thereby challenging the interpretation of attention as a faithful explanation of model behavior. In this context, we investigate whether enforcing functional similarity through logit-based knowledge distillation is sufficient to induce similarity in the internal representations and attention mechanisms of teacher and student models, beyond what a non-distilled model of the same reduced architecture would already exhibit.

The study combines two complementary levels of analysis. First, we examine properties of internal representations through techniques applied to the model's attention maps. To this end, attention similarity metrics are employed, including cosine similarity, Jensen–Shannon (JS) divergence, and matrix norms, both at the layer-averaged level and at the level of individual attention heads. In addition, a spectral analysis of the attention maps is performed to characterize and compare their structural properties, and linear Centered Kernel Alignment (CKA) is used to compare hidden-state representations directly, beyond attention alone.

Second, we investigate whether the decisions produced by the student model can be explained in a manner similar to those of the teacher by using post-hoc XAI techniques such as **LIME** and **SHAP**, comparing the relevance assigned to different input tokens and features. Gradient-based approaches, including **Integrated Gradients** with a zero-vector baseline, are also employed to study the sensitivity of model predictions with respect to the input. Rather than relying solely on qualitative single-example comparisons, we complement these with systematic, aggregate agreement metrics computed over a held-out sample, quantifying top-k attribution overlap, rank and value correlation, sign agreement, deletion-based faithfulness, and explanation stability.

The generated explanations are primarily intended for the technical analysis of models by researchers and practitioners, with the goal of understanding which internal properties emerge under knowledge distillation beyond predictive performance alone. In this way, the present work seeks to provide empirical evidence regarding the extent to which preserving output distributions induces the preservation of internal representations and explanatory mechanisms in compressed models. The full analysis is first conducted as a pilot study on binary IMDb sentiment classification, and then, as the main experiment, extended to the four-class, non-binary AG News topic classification task.

In summary, this work makes the following contributions:

- We design a controlled distillation setting for autoregressive generative classification, compressing a TinyLlama-1.1B teacher by depth into randomly initialized 11-layer students, enabling a direct layer-wise correspondence between teacher and student attention blocks.
- We introduce a family of four student variants sharing a single parameterized loss function but spanning a non-distilled baseline, a weakly-distilled variant, a standard global-KD variant, and a class-restricted KD variant that explicitly addresses the failure of standard vocabulary-level distillation to preserve the argmax relevant to the classification task – allowing us to disentangle the effect of distillation strength from that of depth reduction alone.
- We propose a multi-level representational comparison framework, combining layer-wise and global attention analyses (cosine similarity, Frobenius distance, Jensen–Shannon divergence, spectral effective-rank measures), per-head attention similarity, and linear CKA on hidden states, to quantify structural similarity between teacher and student models beyond simple attention averages.
- We move from qualitative to systematic explanation comparison, evaluating **LIME**, **SHAP**, and **Integrated Gradients** with aggregate top-k overlap, rank/value correlation, sign agreement, faithfulness, and stability metrics, providing quantitative evidence on whether the functional similarity induced by distillation extends to explanatory consistency.
- We extend the full analysis from an initial binary pilot study (IMDb) to a four-class, non-binary main study (AG News), testing whether the observed patterns generalize beyond a single, simple classification task.

II. Related Work

A. Knowledge Distillation

Knowledge distillation (KD) is a model compression technique whose objective is to transfer information from a high-capacity model, referred to as the teacher, to a smaller or more efficient model, referred to as the student [6]. In its traditional formulation, the student is trained not only on the ground-truth labels of the dataset, but also on the probability distributions produced by the teacher. These distributions contain additional information about relationships between classes, model uncertainty, and implicit semantic similarities that conventional discrete labels do not provide. Knowledge transfer is typically implemented through a combination of a standard *cross-entropy* term and a Kullback–Leibler (KL) divergence term that measures the discrepancy between the student and teacher distributions, often softened with a distillation temperature T so that relationships among non-target classes become more visible to the student [6].

KD has been widely employed in deep neural networks and, more recently, in large language models, where the computational costs associated with training and inference are particularly high. However, existing surveys and applications evaluate the success of distillation almost exclusively in terms of functional metrics, such as accuracy or loss on evaluation datasets [5], leaving open whether functional similarity between teacher and student coincides with similarity in their internal representations or in the mechanisms underlying their decisions. In generative language models, classification tasks are additionally often reformulated through *generative classification* [3], in which the target classes correspond to single tokens of the model's own vocabulary rather than to a dedicated classification head. This framing, which we adopt in this work, preserves the model's original architecture and therefore allows internal representations to be compared directly between teacher and student, without the architectural mismatches that a task-specific classification head would introduce.

B. Explainability of Transformer Models

Post-hoc explainability methods aim to interpret the decisions of already-trained, complex models without modifying their architecture. Model-agnostic approaches such as **LIME** [14] and **SHAP** [10] treat the model as a black box: **LIME** locally approximates its behavior with a simpler surrogate model fitted on perturbations of the input, while **SHAP** assigns an individual contribution to each feature through Shapley values derived from cooperative game theory. Gradient-based methods instead exploit the differentiability of neural networks. Simple *saliency maps* estimate sensitivity from raw gradients, but these can be unstable and suffer from saturation; **Integrated Gradients** [16] addresses this limitation by integrating gradients along a path between a reference baseline and the actual input. In natural language processing, these three families of methods are commonly used to quantify the contribution of individual tokens or words to a model's prediction.

Beyond post-hoc attribution, Transformer models expose a potentially interpretable internal mechanism through their attention maps, which quantify how each token distributes its attention over other positions in the sequence. Whether these maps constitute a faithful explanation of model behavior remains contested. Jain and Wallace [7] show that alternative attention configurations can induce similar predictions, challenging the use of attention as a causal explanation, whereas Wiegreffe and Pinter [18] argue that attention can still provide plausible explanations under appropriate testing conditions. To move beyond the interpretation of individual attention matrices, **Attention Rollout** [1] models the cumulative propagation of information across layers, successively combining attention patterns while accounting for residual connections, and thereby provides a more global characterization of how information flows from input tokens to later representations.

Taken together, this body of work has developed largely independent tools: methods for probing a single model's predictions (**LIME**, **SHAP**, **Integrated Gradients**) and methods for probing its internal mechanisms (attention maps, **Attention Rollout**), while the distillation literature has focused on whether a student reproduces the teacher's outputs. Whether a student model that closely reproduces a teacher's predictions also reproduces the internal representations and explanatory mechanisms probed by these XAI tools remains, to the best of our knowledge, unexplored. This work addresses that gap by jointly evaluating representational and explanatory consistency between a teacher and its distilled students.

III. Experiments

A. Experimental Setup

The project was developed in **Python**, due to its widespread adoption within the scientific community and the availability of specialized tools for deep learning and XAI techniques. Model implementation and training were carried out using **PyTorch**, together with Hugging Face's **transformers** library for loading and manipulating language models. In addition, **NumPy** and **Pandas** were used for data processing and manipulation, **Matplotlib** for result visualization, **SciPy** for the rank- and value-correlation statistics used in the aggregate explanation metrics (Section III-B), and the **lime** and **shap** libraries for explanatory analysis of model predictions. Finally, **scikit-learn** was employed to compute performance metrics and evaluate the models on the **test** dataset.

The teacher model used in this work was **TinyLlama-1.1B-Chat-v1.0**. This model is an autoregressive Transformer decoder-only architecture [17] composed of 22 Transformer layers. Its design aims to preserve a structure similar to larger models from the LLaMA family while significantly reducing the number of parameters and the computational cost associated with training and inference. The model incorporates modern attention mechanisms, RMSNorm normalization, and rotary positional embeddings, following an architecture representative of contemporary language models [19]. The same pretrained teacher checkpoint is fine-tuned independently for each of the two datasets studied in this work (Sections III-A1 and III-A2).

To maintain comparability between the internal representations of the teacher and student models, every student architecture used in this work preserves the same embedding configuration, attention mechanisms, and hidden dimensions as the teacher, reducing only the depth of the network. Specifically, the number of Transformer blocks was reduced from 22 to 11, resulting in a $2:1$ correspondence between teacher and student blocks and facilitating layer-wise comparisons. This reduced architecture is shared identically by all four student variants introduced below (Section III-A3); they differ from one another only in their training loss.

When using a generative language model for classification, several challenges arise. First, the natural output of the model corresponds to a distribution over the entire vocabulary, which contains 32000 tokens in the selected teacher model, whereas the task requires only a small, fixed number of classes. Furthermore, generated tokens do not necessarily coincide with the expected class labels. To address this issue, we verified for each dataset that its classification labels could be represented by a single token in the model vocabulary. During training, the loss function was computed over the complete vocabulary distribution, preserving the model's original autoregressive behavior. However, for classification and evaluation purposes, only the logits associated with the target tokens were considered, and the prediction was interpreted as the class with the highest relative probability among them, regardless of whether these tokens were the most probable tokens in the full vocabulary [3].

When working with autoregressive models, the classification problem must be adapted to the generative paradigm used by the model. To achieve this, a base prompt containing a placeholder for the dataset input text was defined, while the expected output consisted of the same prompt concatenated with the target class label. During training, a loss mask was applied so that only the target class token contributed to the loss function. In this way, the model was trained under a

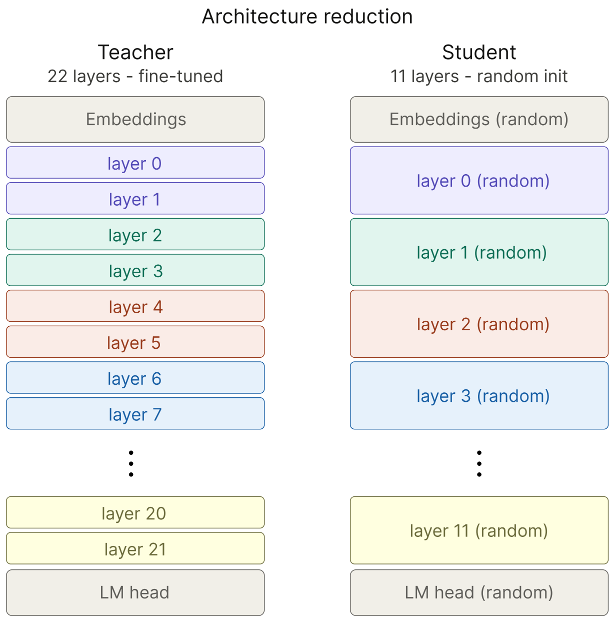


Figure 1: Comparison between the teacher model architecture (left) and the student model architecture (right). The student is initialized entirely with random weights and inherits no parameters from the teacher. Each internal layer of the student has the same size and architecture as the corresponding layers of the teacher.

generative classification framework restricted to a single target output. This framework, along with all training and evaluation procedures described in the remainder of this subsection, is shared identically by both datasets studied in this work; the specifics of each dataset are described in Sections III-A1 and III-A2.

The teacher model corresponds to a language model pre-trained for text generation tasks [19]. However, the task considered in this work differs from its original objective. For this reason, a fine-tuning process was performed independently for each dataset, using a single epoch of the corresponding **train** set, with a **learning rate** of 2×10^{-5}, batch size equal to 1, and the **AdamW** optimization algorithm. All remaining optimizer parameters were left at their default values. Cross-entropy loss computed over the complete vocabulary was used as the training objective. Fine-tuning was performed over all model parameters.

To avoid overfitting and unnecessary training, **early stopping** was applied, halting training if model performance failed to improve after 5 consecutive validation rounds. Validation was performed on random subsets of size 100 sampled from the **validation** dataset. During each validation step, training checkpoints were stored to enable recovery from failures, together with the model weights corresponding to the lowest validation loss observed up to that point.

Every student model is randomly initialized and trained for 3 epochs on the **train** set without applying **early stopping**. The same batch size and optimization algorithm used for the teacher were retained, while the **learning rate** was increased to 1×10^{-4} because training started from random initialization. The **validation** set was used solely for checkpointing and for storing the weights corresponding to the best-performing model observed during training.

1) Setup: IMDb:

```
Review: {input}
Summary:
```

For this study, we used the IMDb sentiment analysis dataset, which consists of 50000 balanced movie reviews labeled with positive or negative sentiment [11], represented in the model's vocabulary by the single tokens *positive* and *negative*. Since the teacher model has a maximum context window of 2048 tokens, the dataset inputs were filtered to ensure that the prompt concatenated with the target label did not exceed this limit. The filtered dataset (49987 valid examples) was then split into **train**, **validation**, and **test** sets, resulting in final sizes of 22492, 2500, and 24995 examples, respectively.

2) Setup: AG News: To test whether the findings obtained on IMDb generalize beyond a simple binary sentiment task, we additionally consider **AG News** [20], a four-class news-topic classification dataset with classes *World*, *Sports*, *Business*, and *Sci/Tech*. As with IMDb, each class label must be representable by a single vocabulary token; the labels *World*, *Sports*, and *Business* already satisfy this, but the tokenizer splits *Sci/Tech* into four sub-tokens (`_Sci`, `/`, `T`, `ech`). We therefore relabel this class *Science*, which tokenizes to a single token, purely to preserve the single-token generative-classification framing used throughout this work; no other change is made to the underlying data. The prompt template used is

```
Article: {input}

Theme:
```

analogous to the IMDb template above. The dataset is class-balanced (30000 articles per class in the original training split); we subsample 7500 examples per class for training (30000 total), 25 per class for validation (100 total), and use the full, balanced test split (1900 per class, 7600 total). As with IMDb, inputs are filtered to fit within the teacher's 2048-token context window; given the shorter length of AG News articles, this filtering removes a negligible number of examples. All remaining training and evaluation hyperparameters (optimizer, learning rates, epochs, early-stopping criterion) are identical to those used for IMDb.

3) A Family of Distillation Losses: The loss function combined a supervised cross-entropy term with a distillation term based on the KL divergence between the teacher and student probability distributions, encouraging both task learning and functional similarity to the teacher. Rather than considering a single student, we train a family of four student variants that share this general parameterized loss but differ in how strongly the distillation terms are weighted, in order to disentangle the effect of the distillation objective from that of depth reduction alone. The general loss is defined as

$$\mathcal{L} = \alpha\,\mathcal{L}_{CE} + (1-\alpha)\,T^2 D_{KL}(P_P \| P_S) + \beta\,T^2 D_{KL}(P_P^{(c)} \| P_S^{(c)})$$

where $\mathcal{L}_{CE}$ denotes the supervised cross-entropy term, $D_{KL}(P_P \| P_S)$ represents the KL divergence between the full-vocabulary probability distributions generated by the teacher (P_P) and the student (P_S), $P_P^{(c)}$ and $P_S^{(c)}$ denote the probability distributions restricted to the classification-relevant tokens (e.g. *positive*/*negative* for IMDb, the four AG News class tokens), and T is the distillation temperature, fixed at $T = 2$ throughout (with the distillation terms additionally scaled by $T^2 = 4$ to preserve gradient magnitudes, following [6]). The coefficients α and β control, respectively, the balance between supervised and global-distillation learning, and the additional weight given to a class-restricted distillation term. Table I summarizes the four student variants considered, together with the teacher.

Table I: The family of models compared in this work. All four students share the same reduced (11-layer) random-initialization architecture and differ only in (α, β).

Model	Init.	α	β	Description
P (teacher)	pretrained	–	–	fine-tuned, CE only
B (baseline)	random	1	0	no distillation
S_{bad}	random	0.9	0	weak global KD
S_1	random	0.5	0	standard global KD
S_2	random	0.5	0.25	+ class-restricted KD

B is a non-distilled baseline: it is trained with cross-entropy only ($\alpha = 1$), with no access to the teacher's outputs during training, and serves to isolate how much of any observed teacher-student similarity is attributable to distillation itself rather than simply to training a smaller model of the same reduced architecture on the same task – the baseline explicitly requested to disentangle these two effects. S_{bad} is a weakly-distilled variant ($\alpha = 0.9$): 90% of its loss is supervised cross-entropy and only 10% is global KD, providing an intermediate point between B and standard distillation. S_1 and S_2 are the original two distilled students: S_1 uses only the global KD term ($\alpha = 0.5, \beta = 0$), while S_2 additionally incorporates the class-restricted term described below ($\beta = 0.25$).

A limitation emerged from the original distillation objective used by S_1. The KL divergence between teacher and student is a global measure over the entire vocabulary distribution. As a consequence, two models may exhibit similar distributions overall without necessarily preserving the relative probability ordering within a specific subset of tokens. In the context of this work, this implies that the most probable class token may differ between teacher and student even when the global distributions are highly similar. To mitigate this issue, S_2 additionally incorporates the class-restricted KL term $\beta\,T^2 D_{KL}(P_P^{(c)} \| P_S^{(c)})$, computed exclusively over the logits associated with the class tokens and weighted more lightly than the global term so as to act as a complementary regularizer rather than override the distributional information captured by global distillation. Consequently, S_2's optimization objective seeks not only to approximate the full vocabulary distributions between teacher and student, but also to preserve the relative probability ordering of the classes relevant to the classification task.

The fine-tuned teacher and all four student variants were evaluated on the **test** set of each dataset before comparisons were conducted. This evaluation was performed to verify the correct operation of the models and their expected behavior on the task. Since the problem corresponds to generative classification with a single output token, the models were evaluated using a single forward pass and analyzing the logits of the target classes. This approach is computationally less expensive than using the full generative decoding loop and also avoids generating tokens outside the expected class set. Consequently, the comparison was restricted to the logits associated with the target classes. The final prediction was obtained by selecting the class corresponding to the highest logit among the target tokens. After classifying all examples, the metrics *accuracy*, *precision*, *recall*, *F1-score*, and the confusion matrix were computed.

After verifying the correct operation of the models on the **test** set, comparisons of outputs and internal representations were performed between the teacher and all four student variants. Functional evaluation was conducted using classical post-hoc XAI methods, namely **LIME**, **SHAP**, and **Integrated Gradients**, with the objective of identifying which words or tokens exert the greatest influence on classification decisions. The resulting explanations were then used to compare the degree of explanatory consistency between the teacher and student models, both qualitatively, on one illustrative example per dataset, and systematically, using the aggregate agreement and faithfulness metrics defined in Section III-B, computed over a held-out sample of $N = 100$ test examples per dataset (balanced across classes, fixed random seed).

For **LIME**, the **LimeTextExplainer** was used, generating random perturbations of the input text by removing complete words and observing the resulting variations. For **SHAP**, text masking was employed, producing a similar procedure based on hiding words from the input text. **Integrated Gradients** was implemented using a zero embedding vector as the baseline and analyzing how different tokens affected the output. Formally, given an input x, a baseline x', and the model function $F(\cdot)$ whose prediction is to be explained, **Integrated Gradients** attributes to each input dimension i the quantity

$$\begin{aligned}\mathrm{IG}_i(x) &= (x_i - x_i') \int_0^1 \frac{\partial F\left(x' + \alpha(x - x')\right)}{\partial x_i} d\alpha \\ &\approx (x_i - x_i') \frac{1}{m} \sum_{k=1}^{m} \frac{\partial F\left(x' + \frac{k}{m}(x - x')\right)}{\partial x_i}\end{aligned}$$

where $\alpha \in [0, 1]$ parameterizes the interpolation path between x' and x, and m denotes the number of steps used to approximate the integral in practice [16]. Attributions are computed with respect to the input embeddings; for consistency with the other methods, token-level attribution scores were aggregated after reconstructing complete words.

The output probability distributions of the teacher and each student model were compared at temperatures $T = 1$ and $T = 2$, considering only the global vocabulary distributions. These temperatures were selected because $T = 1$ corresponds to standard inference and $T = 2$ was used during distillation training. The objective of this analysis was to assess the probabilistic similarity of the predictions beyond the final predicted label. In this way, the study evaluates the extent to which each model exhibits similar distributional behavior to the teacher, even in cases where their point predictions differ.

Comparisons of internal representations focused on similarities and differences in attention mechanisms and hidden states, beyond the importance assigned to individual tokens. Experiments were conducted using three complementary perspectives: a microscopic (granular) analysis, a macroscopic (global) analysis, and a set of finer-grained probes (per-head attention and hidden-state comparisons). The granular approach compared teacher attention blocks with potentially corresponding layers in each student model. More specifically, attention map i of each student was compared with the average attention map obtained from teacher layers $2i$ and $2i + 1$; the corresponding attention-rollout comparison is discussed alongside the global comparisons below, where it plays a more prominent role. This comparison is enabled by the $2 : 1$ architectural relationship between teacher and student models. The global approach focused on comparing the average attention maps and attention rollouts of the complete models. All comparisons were performed on attention maps averaged over held-out samples from the **test** dataset rather than on individual examples ($N = 200$ examples for IMDb, $N = 200$ for AG News). At the representational level, the goal was to characterize the average behavior of the models regardless of the analysis scale. Additional comparisons between student models were also conducted to identify common structural patterns. To probe representational similarity at a finer grain than layer-averaged attention, we additionally compare individual attention heads directly (rather than head-averaged maps), over a smaller sample of $N = 50$ prompts, and compare full hidden-state representations (rather than only attention) across all layers via linear CKA, over $N = 100$ prompts. The specific metrics used to quantify all of these comparisons, together with their formal definitions, are described in Section III-B.

The experiments were conducted using an **NVIDIA GeForce RTX 5070 Ti** GPU with 16 GB of **GDDR7** VRAM. The repository containing the source code and experimental results is available at **https://github.com/PseudoKiwi/ProyectoXAI**.

The writing of this manuscript was partially assisted by generative language models used exclusively for stylistic revision and textual reformulation tasks: Claude Sonnet 4.6 [2] (Anthropic) and ChatGPT [12] (OpenAI).

B. Comparison Metrics

Attention maps can be interpreted as structured representations of token dependencies, where each row describes a probability distribution over the attended context. This makes it possible to compare representations across models using geometric, matrix-based, and probabilistic metrics in order to quantify structural similarities and differences in the learned attention patterns. In this work, we employ complementary measures that capture different properties of attention maps, including global alignment, distributional discrepancies, and spectral complexity.

1) Aggregated Attention Representations: P_{avg} and P_r: For each attention head and layer, self-attention mechanisms produce row-stochastic matrices that quantify dependency relationships between tokens. In this work, two complementary aggregations of these matrices are used to characterize a model's attention behavior at the layer and global levels. The *average attention map*, denoted P_{avg}, is obtained by averaging attention matrices over heads and over dataset examples. **Attention Rollout** [1], denoted P_r, instead models the cumulative propagation of information across layers by successively combining attention matrices while accounting for residual connections, thereby providing a more global approximation of how information flows from input tokens to later representations than any individual attention map or simple average can capture.

2) Cosine Similarity: Cosine similarity is a widely used measure for comparing vector representations, quantifying the degree of angular alignment between two objects independently of their magnitude. Given two vectorized representations A and B, cosine similarity is defined as

$$\cos(\theta) = \frac{\langle A, B \rangle}{\|A\|_2 \|B\|_2}$$

where θ denotes the angle between the vectorized representations A and B, $\langle \cdot, \cdot \rangle$ denotes the inner product, and $| \cdot |_2$ denotes the Euclidean norm. By definition, its value is bounded between -1 and 1.

When applied to attention maps, this metric enables the evaluation of structural similarity between attention patterns learned by different models. It is particularly useful for identifying global alignment between representations, even when differences in the scale of the assigned probabilities are present.

3) Frobenius Distance: The Frobenius distance quantifies absolute discrepancies between matrices by considering element-wise differences. Given two attention matrices A and B, it is defined as

$$d_F(A, B) = \|A - B\|_F = \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

This metric can be interpreted as a matrix generalization of Euclidean distance and is particularly useful for measuring the extent to which two attention maps differ in absolute terms. Unlike cosine similarity, the Frobenius distance is sensitive to variations in scale and magnitude, capturing local differences that may not be reflected solely through geometric alignment. In the context of this work, this measure enables the evaluation of the proximity between attention maps generated by teacher and student models by quantifying global differences in the distribution of attention weights.

Since attention maps are row-stochastic matrices of dimension $n \times n$, the Frobenius distance between two such matrices admits a theoretical upper bound. For each row i, the entries A_{ij} and B_{ij} are non-negative and sum to 1.

$$\sum_j (A_{ij} - B_{ij})^2 \leq \sum_j A_{ij}^2 + \sum_j B_{ij}^2 \leq \sum_j A_{ij} + \sum_j B_{ij} = 2$$

where we used the fact that $x^2 \leq x$ for $x \in [0,1]$. Summing over the n rows yields

$$d_F(A,B)^2 = \sum_{i,j} (A_{ij} - B_{ij})^2 \leq 2n$$

and therefore

$$d_F(A,B) \leq \sqrt{2n}$$

This bound is attained when each row of A and B concentrates all its probability mass on different tokens. For $n \times n$ attention maps with $n = 2048$, the maximum possible value is $\sqrt{2 \cdot 2048} = 64$.

4) Jensen-Shannon Divergence: Since the rows of attention maps can be interpreted as probability distributions over tokens, it is natural to employ information-theoretic measures to compare representations. In this work, we use the Jensen–Shannon (JS) divergence, a symmetric and bounded measure derived from the KL divergence.

For two probability distributions P and Q, the JS divergence is defined as

$$JS(P,Q) = \frac{1}{2} D_{KL}(P \| M) + \frac{1}{2} D_{KL}(Q \| M)$$

where

$$M = \frac{1}{2}(P + Q)$$

denotes the average distribution of P and Q.

Unlike the Kullback–Leibler divergence, JS is symmetric and bounded, taking values in the interval $[0, \ln 2]$ when measured in nats, where 0 corresponds to identical distributions and $\ln 2$ corresponds to distributions with disjoint support. These properties make it particularly suitable for comparing attention distributions [9].

5) Effective Ranks: Beyond direct comparisons between attention maps, global properties of representations can be characterized through spectral analysis. Using decompositions such as the *Singular Value Decomposition* (SVD), the singular values associated with attention matrices can be analyzed, making it possible to study how information is distributed across different directions of the representational space.

In this context, the concept of *effective rank* provides a measure of the functional dimensionality of a representation, taking into account not only the algebraic rank of the matrix but also the relative concentration of energy in its principal components. Rather than assuming that all dimensions contribute equally, these measures estimate how many spectral directions participate significantly in the representation.

To quantify the geometric complexity of attention maps, two complementary definitions of effective rank were employed. The first corresponds to the entropy-based effective rank (*Entropy Rank*), defined in terms of the Shannon entropy of the normalized singular values [15].

$$ER = \exp\left(- \sum_i p_i \log p_i \right), \qquad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

where σ_i denotes the i-th singular value of the matrix.

Additionally, an energy-based definition known as the *Participation Ratio* (PR) was employed [13].

$$PR = \frac{\left(\sum_i \sigma_i^2\right)^2}{\sum_i \sigma_i^4}$$

This metric estimates how many spectral components contribute significantly to the representation while penalizing distributions that are highly concentrated in a small number of dominant directions. The combined use of both measures enables the analysis of similarities and differences between representations from complementary perspectives on the structural complexity of attention mechanisms.

Beyond the layer-averaged attention maps considered above, the same cosine-similarity, Frobenius-distance, and JS-divergence formulas are additionally applied at the level of individual attention heads (Section IV-A5), simply by treating each head's attention pattern as its own vector, without requiring any new formal definition.

6) Linear Centered Kernel Alignment (CKA): While the metrics above operate on attention maps, hidden-state representations can be compared directly using *Linear Centered Kernel Alignment* (CKA) [8], a similarity measure between two sets of representations that is invariant to orthogonal transformations and isotropic scaling of either representation, making it well suited to comparing hidden states from models of different width or trained under different objectives. Given two matrices of representations $X \in \mathbb{R}^{n \times d_1}$ and $Y \in \mathbb{R}^{n \times d_2}$, obtained from the same n input examples, let $K = XX^\top$ and $L = YY^\top$ denote their Gram matrices. Linear CKA is defined as

$$\mathrm{CKA}(K,L) = \frac{\mathrm{HSIC}(K,L)}{\sqrt{\mathrm{HSIC}(K,K)\,\mathrm{HSIC}(L,L)}}$$

where HSIC denotes the (empirical) Hilbert–Schmidt Independence Criterion, computed on the doubly-centered Gram matrices. CKA is bounded in $[0,1]$, with 1 indicating identical representational geometry up to rotation and scale. In this work, X and Y are the last-real-token hidden states extracted at every layer (including the embedding layer) of the teacher and of a given student, respectively, over a fixed sample of prompts; comparing all teacher layers against all student layers, rather than only matched layer pairs, yields a full teacher-layer $\times$ student-layer CKA matrix (Section IV-A6).

7) Explanation Agreement and Faithfulness Metrics: To move beyond the qualitative comparison of individual LIME, SHAP, and Integrated Gradients explanations, we additionally compute a set of aggregate metrics over a held-out sample of examples, quantifying the agreement between a given student's word-level attribution scores and the teacher's, as well as the intrinsic quality of each explanation. Let s_a, s_b denote the word-to-score attribution dictionaries produced by the same method for two models on the same input, and let K be the set of words attributed by both (their common support).

Top-k Jaccard overlap. Given k (fixed to $k = 10$ throughout), let $T_a, T_b \subset K$ be the top-k words by $|s_a(\cdot)|$ and $|s_b(\cdot)|$, respectively. The overlap is

$$J_k(s_a, s_b) = \frac{|T_a \cap T_b|}{|T_a \cup T_b|}$$

undefined (and excluded from averages) when $|K| < k$.

Rank and value correlation. The Spearman rank correlation ρ and Pearson correlation r between $\{s_a(w)\}_{w\in K}$ and $\{s_b(w)\}_{w\in K}$ quantify, respectively, the agreement in the relative ordering and in the linear relationship of the signed attribution scores over the common support.

Sign agreement@k. The fraction of T_a (the top-k words by $|s_a(\cdot)|$) for which $\mathrm{sign}(s_a(w)) = \mathrm{sign}(s_b(w))$, measuring whether the two models agree on the polarity (supporting vs. contradicting the predicted class) of the most important words, independently of magnitude.

Faithfulness@k (deletion). For a single model with attribution s on input x, let x_{-k} be x with its top-k $|s(\cdot)|$ words masked, and let $p(\cdot \mid x)$ be the model's predicted probability of the true class. Faithfulness is defined as

$$F_k(s, x) = p(y \mid x) - p(y \mid x_{-k})$$

a large positive value indicating that the words the method deems most important are indeed responsible for a large share of the model's confidence in the correct class [4]. Unlike the three metrics above, faithfulness is computed per model (against its own predictions), not between a student and the teacher, and is therefore also reported for the teacher itself as a reference point.

Stability. For a given model and method, explanations are recomputed M times with different random seeds (for LIME, different perturbation samples) on the same subset of examples, and stability is reported as the mean pairwise top-k Jaccard overlap (J_k, defined above) across all $\binom{M}{2}$ seed pairs, averaged over examples. Unlike the agreement metrics above, stability characterizes a single model's explanations in isolation, independent of the teacher.

IV. Results

A. IMDb: Pilot Study

This section reports the pilot study conducted on IMDb sentiment classification, covering all five models of Table I (P, B, S_{bad}, S_1, S_2). Given its role as an initial, fast-turnaround study, results are reported compactly, with full per-layer detail retained where it is cheap to do so and cross-model comparisons summarized in prose where a full pairwise table would add little beyond what the teacher-referenced comparisons already show; the main study, conducted on the more complex AG News task, is reported in full depth in Section IV-B.

1) Compression: The teacher model contains a total of $|P| = 1100048384$ parameters, whereas each student, constructed with half the depth, contains $|S| = 615561216$ parameters. This corresponds to a *compression rate*

$$CR = \frac{|P|}{|S|} \approx 1.787$$

which is equivalent to an approximate 44% reduction in the number of parameters relative to the teacher model. The parameter reduction is not exactly proportional to the reduction in depth, since components independent of the number of layers, such as the vocabulary embeddings and the output layer (*lm head*), were kept identical between teacher and students in order to preserve representational comparability. This compression rate is identical for all four student variants, since they share the same reduced architecture and differ only in their training loss (Table I).

*2) Evaluation on the **Test** Set:* Table II reports classification metrics on the **test** set for all five models. The teacher achieves near-perfect performance, with only 798 out of 24995 examples misclassified. B, the non-distilled baseline, reaches a moderate 0.8575 accuracy (confusion matrix $\left[\begin{smallmatrix}10405 & 2090\\ 1473 & 11027\end{smallmatrix}\right]$, rows/columns ordered positive/negative), noticeably below the teacher but well above chance. S_{bad}, the weakly-distilled variant, performs worse still (0.8130 accuracy, confusion matrix $\left[\begin{smallmatrix}8647 & 3848\\ 827 & 11673\end{smallmatrix}\right]$), with a bias toward the negative class reminiscent of, though less extreme than, S_1's. S_1, trained under a standard global-KD objective, performs worst of all five models (0.6718 accuracy), with a strong bias toward the negative class (7997 of 12495 positive examples misclassified as negative; Figure 3). S_2, which adds the class-restricted KD term, recovers most of this gap and approaches the teacher's performance (0.9343 accuracy, 1643 misclassified), with a mild residual bias toward the positive class (Figure 4; $\approx 69\%$ of its errors are false positives).

Table II: Performance metrics on the generative classification task, evaluated on the **test** set, for all five IMDb models.

Metric	P	B	S_{bad}	S_1	S_2
Accuracy	0.9681	0.8575	0.8130	0.6718	0.9343
Precision	0.9681	0.8583	0.8324	0.7810	0.9354
Recall	0.9681	0.8574	0.8129	0.6718	0.9343
F1 score	0.9681	0.8574	0.8102	0.6364	0.9342

Average KL divergence to the teacher was computed for all four student variants at temperatures $T = 1$ and $T = 2$ over a fixed, balanced held-out subset ($N = 3000$); these values supersede the ones reported in earlier iterations of this study, which used an inconsistent reduction across models. Table III reports the results. B is, by a wide margin, the least faithful to the teacher's output distribution, while all three distilled variants remain close to the teacher; notably, S_1 attains a *lower* KL than S_2 at both temperatures, despite S_2's far better classification performance (Table II) – global KL divergence is consequently not a predictive proxy for classification performance on this task, and, as elaborated in Section IV-A3, nor is it a predictive proxy for explanatory alignment. The strong class bias observed in S_1's confusion matrix reflects the fact that global KL divergence, used as its sole distillation objective, does not explicitly preserve the argmax among the target-class logits; S_2's additional class-restricted term directly targets this failure mode and substantially improves classification performance, at the cost of a higher global KL divergence.

Table III: Average KL divergence to the teacher P, computed over a fixed balanced subset ($N = 3000$) of the **test** set, at temperatures $T = 1$ and $T = 2$.

	B	S_{bad}	S_1	S_2
$KL_{T=1}$	2.8473	0.1037	0.1042	0.1110
$KL_{T=2}$	1.6861	0.1317	0.1274	0.1926

3) Functional Evaluation and Explanation Comparison: Figure 5 illustrates **LIME** explanations for one representative negative-sentiment example, across all five models. The teacher assigns negative importance to genuinely negative-sentiment words (e.g. *predictable*, *wooden*, *dull*, *uninspiring*) and positive importance to positive-sentiment words, consistent with the ground truth; since P is a general-purpose pretrained model [19], this is expected, as the tokens most relevant to a sentiment task should be those that genuinely convey sentiment. All four students, in contrast, assign their highest importance to semantically empty function words (*and*, *the*) regardless of their classification performance – this pattern is visually indistinguishable between B, S_{bad}, S_1, and S_2 in

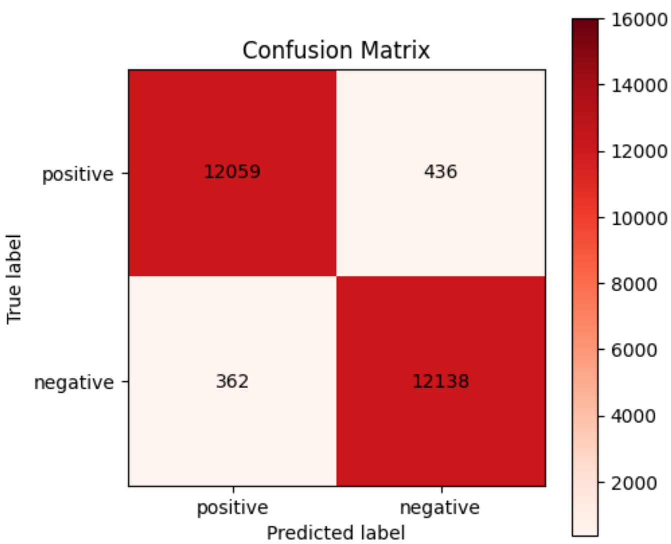


Figure 2: Confusion matrix of the teacher model on the generative classification task after fine-tuning, evaluated on the **test** dataset.

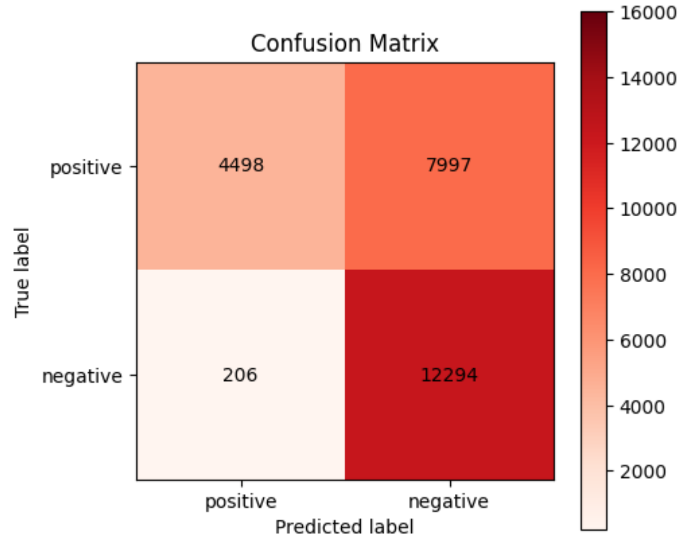


Figure 3: Confusion matrix of S_1 on the generative classification task, evaluated on the **test** dataset.

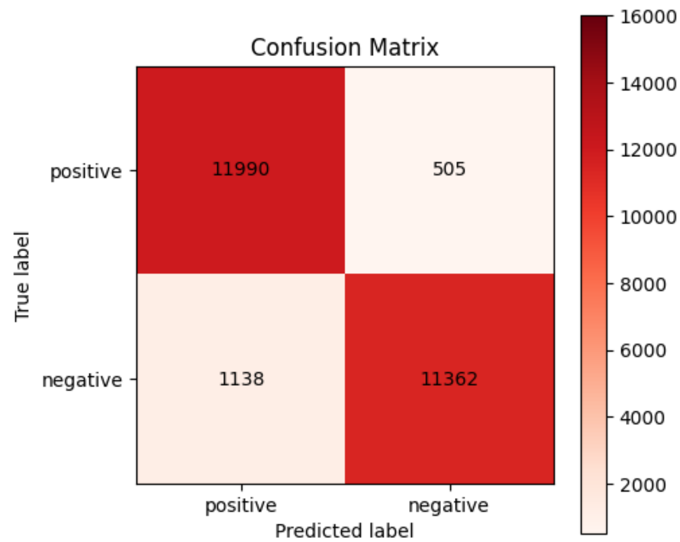
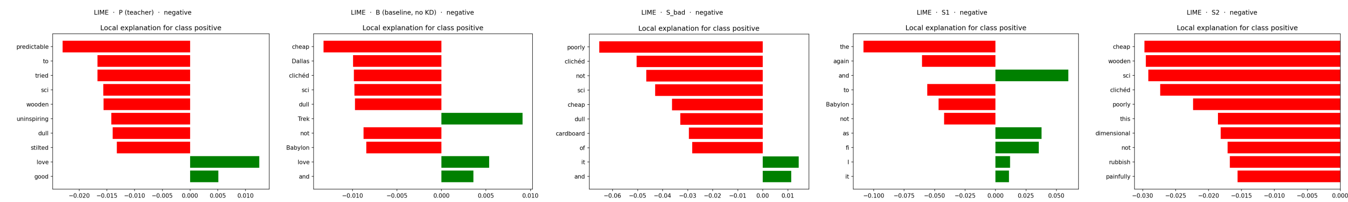


Figure 4: Confusion matrix of S_2 on the generative classification task, evaluated on the **test** dataset.

this example, and the same qualitative behavior is observed in the corresponding **SHAP** explanations (not shown). This qualitative impression is confirmed systematically below.

Beyond this single example, we compute the aggregate agreement and faithfulness metrics of Section III-B7 over a held-out sample of $N = 100$ test examples, comparing each student's attributions against the teacher's. Tables IV, V, and VI report the results for **LIME**, **SHAP**, and **Integrated Gradients**, respectively (with a zero embedding vector used as the IG baseline, in place of the padding token used in earlier iterations of this study).

For both **LIME** and **SHAP**, S_2 is the best-aligned student with the teacher on every agreement metric and the most



(a) P (b) B (c) S_{bad} (d) S_1 (e) S_2

Figure 5: **LIME** explanations for a representative negative-sentiment IMDb example, across all five models, computed using 5000 random word-removal perturbations of the input text.

Table IV: **LIME** aggregate agreement metrics (vs. teacher), $N = 100$, $k = 10$. P's faithfulness is reported alone, as a self-comparison reference.

Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.172
B	0.183	0.248	0.359	0.775	0.083
S_{bad}	0.170	0.216	0.329	0.770	0.202
S_1	0.136	0.189	0.220	0.731	0.177
S_2	0.212	0.299	0.415	0.796	0.267

Table V: **SHAP** aggregate agreement metrics (vs. teacher), $N = 100$, $k = 10$.

Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.118
B	0.117	0.239	0.283	0.788	0.025
S_{bad}	0.123	0.238	0.293	0.761	0.093
S_1	0.096	0.245	0.208	0.723	0.058
S_2	0.176	0.333	0.387	0.800	0.120

Table VI: **Integrated Gradients** (zero baseline) aggregate agreement metrics (vs. teacher), $N = 100$, $k = 10$.

Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.010
B	0.793	-0.005	0.086	0.526	0.010
S_{bad}	0.778	0.028	0.120	0.538	0.085
S_1	0.814	0.010	-0.021	0.513	0.025
S_2	0.787	0.039	0.077	0.503	0.019

faithful of the four students, followed by S_{bad}, then B and S_1 (which alternate depending on the metric); none of the four students, however, reaches faithfulness values remotely close to being semantically satisfactory, consistent with the qualitative impression from Figure 5. **Integrated Gradients** tells a qualitatively different story: all models, including the teacher, exhibit low faithfulness, and while top-k overlap is high ($\approx$ 0.78–0.81) across all four students, rank and value correlations are close to zero (occasionally negative), and sign agreement hovers near chance ($\approx$ 0.50–0.54). This combination – high token-identity overlap but near-zero rank agreement – indicates that IG tends to select a similar *set* of tokens across models without agreeing on their relative importance or polarity, a pattern we revisit in Section IV-D.

Table VII reports **LIME** stability (mean pairwise Jaccard@10 across 5 re-runs, $N = 20$). The teacher's explanations are less stable than S_2's, and more stable than B's, S_{bad}'s, and S_1's – stability alone is therefore not a proxy for explanation quality, since the teacher's substantially more meaningful explanations (Figure 5) are not the most reproducible ones.

Table VII: **LIME** stability (mean pairwise Jaccard@10 across 5 seeds, $N = 20$).

Model	P	B	S_{bad}	S_1	S_2
Stability	0.548	0.587	0.684	0.611	0.710

Taken together, these results reinforce the point raised in Section IV-A2: performance, distributional similarity, and explanatory alignment are distinct axes that need not covary. S_2 is simultaneously the best-performing student (Table II) and the best-aligned with the teacher on **LIME/SHAP** agreement and faithfulness, yet it has a *higher* global KL divergence than S_1 (Table III) – the student closest to the teacher in raw output distribution is neither the most accurate nor the most explanatorily faithful.

4) Representational Comparison: This section presents the results of the comparisons between the attention maps and hidden states of the models. We denote by B the non-distilled baseline, S_{bad} the weakly-distilled student, S_1 the student trained using standard global KD, S_2 the student trained with the additional class-restricted KD term, and P the teacher model, as in Table I. *These numbers, recomputed as part of the corrections underlying this revision, supersede the layer-wise values reported in earlier iterations of this study.* Cross-model comparisons other than each student against the teacher (e.g. B vs. S_1) are summarized in prose rather than tabulated in full, following the same rationale as for the KL comparisons above.

a) Layer-wise Comparisons: For each student attention map i, comparisons are performed against the average attention map obtained from teacher layers $2i$ and $2i + 1$ (P_{avg}); the corresponding rollout-referenced comparisons follow very similar trends and are omitted from the compact tables below for brevity. Figure 6 summarizes the resulting layer-wise trends for all four students at a glance.

Table VIII reports the cosine similarity between each student's attention map and P_{avg}. All four models show a similarly high first-layer similarity (≥ 0.92), which then drops sharply from layer 1 onward. Notably, B – which receives no supervision from the teacher whatsoever – is consistently the *least* similar to the teacher of the four models in the majority of

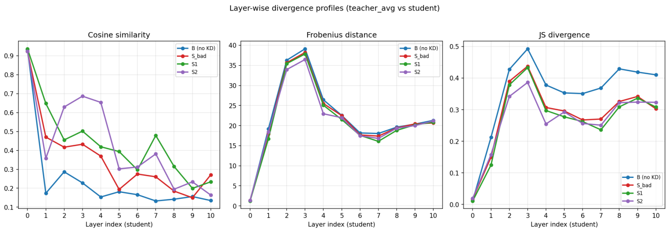


Figure 6: Layer-wise cosine similarity, Frobenius distance, and JS divergence between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i+1$), for all four IMDb students.

layers (e.g. 0.15–0.29 in layers 1–10, versus 0.19–0.69 for the distilled variants), suggesting that on this dataset, distillation does confer some attention-level alignment beyond what depth reduction alone would produce – in contrast with the pattern observed on AG News (Section IV-B3).

Table VIII: Cosine similarity between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i+1$).

i	B	S_{bad}	S_1	S_2
0	0.9330	0.9230	0.9367	0.9233
1	0.1735	0.4703	0.6479	0.3579
2	0.2869	0.4170	0.4552	0.6293
3	0.2283	0.4325	0.5018	0.6865
4	0.1538	0.3698	0.4191	0.6536
5	0.1816	0.1938	0.3938	0.3026
6	0.1659	0.2751	0.2971	0.3121
7	0.1328	0.2604	0.4788	0.3814
8	0.1421	0.1849	0.3150	0.1948
9	0.1561	0.1498	0.1995	0.2338
10	0.1350	0.2709	0.2340	0.1641

Table IX reports the corresponding Frobenius distances. Unlike cosine similarity, the four models are nearly indistinguishable from each other on this metric, all following essentially the same profile: near-perfect alignment at $i = 0$ (distances ≈ 1.2–1.4, i.e. below 2% of the theoretical maximum of 64), rising sharply to a peak around $i = 2$–3 (≈ 34–39, ≈ 55–62% of the bound), then decreasing and stabilizing around 17–21 in later layers. Row-averaged JS divergence (not tabulated) mirrors this same profile for all four models, from ≈ 0.01–0.02 at $i = 0$ up to ≈ 0.35–0.49 at the peak.

Table IX: Frobenius distance between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i+1$).

i	B	S_{bad}	S_1	S_2
0	1.2648	1.3557	1.2356	1.3529
1	19.1492	17.8870	16.7155	18.3989
2	36.2836	35.6589	35.4357	33.9489
3	39.1174	38.2045	37.8417	36.4605
4	26.4354	25.4998	25.0765	22.9474
5	22.5318	22.4848	21.5096	21.9667
6	18.1655	17.6709	17.5652	17.4682
7	18.0399	17.3913	16.0823	16.8076
8	19.6057	19.4096	18.8159	19.3582
9	20.3702	20.4235	20.1939	20.0093
10	21.3099	20.6607	20.8501	21.1837

Table X and Table XI report the spectral effective-rank measures. P_{avg} remains close to rank-one throughout ($PR \approx 1$ beyond $i = 0$), while P_r is highly dispersed ($ER \approx 1650$–2027). All four students occupy an intermediate position, with effective ranks generally increasing toward deeper layers, and no consistent ordering among B, S_{bad}, S_1, and S_2 – e.g. B has the lowest ER at $i = 0$ (60.2) but is squarely in the middle of the pack from $i = 4$ onward.

Table X: Participation Ratio (PR) for each attention map.

i	B	S_{bad}	S_1	S_2	P_{avg}	P_r
0	2.55	2.62	2.45	2.73	3.26	399.64
1	5.08	2.12	1.70	5.04	1.01	1.86
2	3.37	2.98	2.45	1.65	1.00	1.31
3	2.94	3.34	2.48	2.00	1.00	1.29
4	4.96	4.39	4.27	2.39	1.00	1.55
5	5.57	17.15	5.90	7.65	1.01	1.72
6	7.32	11.45	10.63	13.57	1.01	2.14
7	13.72	13.40	7.50	7.13	1.02	2.16
8	11.50	15.95	5.49	14.88	1.01	1.97
9	8.93	11.65	15.77	11.40	1.01	1.90
10	9.68	8.61	17.02	12.54	1.02	1.88

Table XI: Entropy-based effective rank (ER) for each attention map.

i	B	S_{bad}	S_1	S_2	P_{avg}	P_r
0	60.2	116.9	95.6	107.3	309.3	2026.6
1	329.4	113.1	101.8	399.1	42.2	1819.4
2	277.5	298.3	229.6	179.8	5.2	1667.7
3	327.7	398.4	303.6	360.5	7.0	1650.1
4	314.5	437.1	516.8	484.6	20.4	1761.2
5	355.8	507.5	469.8	489.5	26.3	1797.7
6	402.0	497.1	486.0	514.5	66.3	1849.2
7	505.1	554.9	505.2	488.0	84.2	1850.0
8	506.8	521.7	431.3	501.2	64.1	1832.5
9	460.2	514.3	549.1	551.3	45.5	1823.9
10	491.4	508.5	550.6	528.0	129.1	1819.1

Across all four models, S_1 and S_2 remain highly similar to each other in every layer (cosine similarity > 0.81 throughout, not tabulated), consistent with earlier iterations of this study; the same holds for B and S_{bad} against the distilled students, indicating that at the layer-wise level, differences between the loss variants of Table I are second-order compared to the shared effect of the reduced architecture and task.

b) Global Comparisons: Table XII reports comparisons between whole-model aggregated representations. As in earlier iterations of this study, the students' *rollout* representations align strongly with the teacher's global average map ($\cos \geq 0.91$, $d_F \leq 10.6$), markedly higher than between the students' own average maps and P_{avg} ($\cos \approx 0.21$–0.44, $d_F \approx 20$–22); this should still be interpreted cautiously, as both aggregation schemes converge toward more uniform distributions through different mechanisms (Section III-B). All four students, including the non-distilled B, participate in this pattern to a similar degree, so it cannot be attributed specifically to distillation. S_1 and S_2 remain close to each other in both aggregations ($\cos = 0.988$ for average maps, $\cos = 0.998$ for rollouts), and B and S_{bad} are likewise close to the distilled students (e.g. B_{avg}–$S_{1,avg}$: $\cos = 0.714$; $S_{bad,avg}$–$S_{1,avg}$: $\cos = 0.988$).

Table XII: Global attention comparisons (whole-model average and rollout maps) against the teacher's global average (P_{avg}) and rollout (P_r). The cosine similarity for $S_{1,avg}$–P_{avg} could not be recovered due to a logging error in the source notebook (the corresponding Frobenius distance was logged twice instead); this cell is intentionally left blank rather than estimated.

Comparison	Cosine	d_F
B_{avg}–P_{avg}	0.2087	21.64
B_{avg}–P_r	0.1405	44.88
B_r–P_{avg}	0.9750	5.09
B_r–P_r	0.9694	23.78
$S_{bad,avg}$–P_{avg}	0.3168	21.16
$S_{bad,avg}$–P_r	0.2455	44.40
$S_{bad,r}$–P_{avg}	0.9058	10.63
$S_{bad,r}$–P_r	0.8909	32.59
$S_{1,avg}$–P_{avg}	–	20.69
$S_{1,avg}$–P_r	0.3504	43.92
$S_{1,r}$–P_{avg}	0.9069	9.99
$S_{1,r}$–P_r	0.8956	31.37
$S_{2,avg}$–P_{avg}	0.4440	20.54
$S_{2,avg}$–P_r	0.3797	43.77
$S_{2,r}$–P_{avg}	0.9251	9.59
$S_{2,r}$–P_r	0.9133	31.46

Table XIII reports the global spectral analysis. As at the layer-wise level, the teacher's global rollout is close to rank-one ($PR \approx ER \approx 1$), while every student's average map is markedly more dispersed ($ER \approx 284$–464); again, B's dispersion ($ER = 284.3$) is comparable to, rather than clearly lower than, the distilled students' ($ER = 446$–458), so this global spectral signature is not obviously a distillation-specific effect either.

Table XIII: Global spectral analysis (Participation Ratio and entropy-based effective rank).

Representation	PR	ER
P_{avg}	1.0063	40.47
P_r	1.0000	1.00
B_{avg}	3.4352	284.34
B_r	1.0022	1.98
$S_{bad,avg}$	5.8041	457.74
$S_{bad,r}$	1.0104	3.38
$S_{1,avg}$	4.3530	446.50
$S_{1,r}$	1.0057	3.03
$S_{2,avg}$	4.7671	463.61
$S_{2,r}$	1.0052	3.23

5) Per-Head Attention Similarity: Beyond layer-averaged attention, Figure 7 compares individual attention heads directly: for each student, a 32×32 cosine-similarity matrix between the teacher's heads and the student's heads, averaged over the 11 matched layer pairs and over $N = 50$ prompts (Section III-A). All four students show a qualitatively similar, diffuse pattern: similarities are uniformly modest (mostly in the 0.2–0.5 range, with no head pair exceeding ≈ 0.6) and organized into vertical bands, indicating that a subset of student heads is moderately similar to *most* teacher heads rather than any single teacher head being distinctively reproduced by a matching student head. B's heatmap is visibly weaker

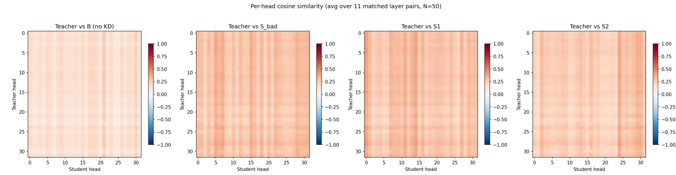


Figure 7: Per-head cosine similarity between teacher and student attention heads, averaged over 11 matched layer pairs and $N = 50$ prompts, for all four IMDb students.

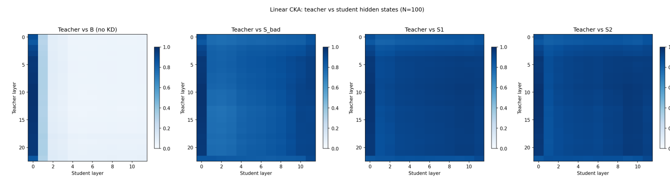


Figure 8: Linear CKA between teacher and student hidden states (all layers, including the embedding layer, $N = 100$ prompts), for all four IMDb students.

overall (paler) than the three distilled variants', suggesting a modest distillation effect at the level of individual heads for this dataset, consistent with the layer-wise cosine-similarity finding above – though the effect is far less pronounced than at the level of full attention maps.

6) Hidden-State Comparison (Linear CKA): Figure 8 reports the full teacher-layer $\times$ student-layer linear CKA matrix (Section III-B6) for each student, computed on last-token hidden states over $N = 100$ prompts. All three distilled students exhibit the same qualitative pattern: near-zero CKA with the embedding layer and the first one or two teacher layers, rising sharply and saturating close to 1 from roughly the middle of the teacher's depth onward, essentially independent of which student layer is being compared – i.e. once the teacher's representations become sufficiently deep, they become globally similar (in the CKA sense) to *every* student layer, not only to the nominally corresponding one. Strikingly, B's CKA heatmap is visually almost indistinguishable from the three distilled students', reaching the same saturation pattern in deep layers. This is the clearest evidence in this work that hidden-state geometric similarity to the teacher, as captured by CKA, is largely insensitive to whether a model was distilled at all, at least on IMDb.

B. AG News: Main Study

We now turn to AG News, the four-class, non-binary classification task introduced in Section III-A2, used as the main experimental study of this work. The same five models (Table I), the same training procedure, and the same metrics as in the IMDb pilot (Section IV-A) are used throughout, now reported in full depth.

*1) Evaluation on the **Test** Set:* Table XIV reports macro-averaged classification metrics on the full 7600-example **test** set. In sharp contrast with IMDb, the non-distilled baseline B achieves the *highest* raw accuracy of all five models, ahead of every distilled variant and the teacher itself: $B > S_{bad} > S_2 > S_1 > P$. This is a central finding of this study and is discussed further in Section IV-C.

Table XIV: Performance metrics (macro-averaged) on the generative classification task, evaluated on the full **test** set (7600 examples), for all five AG News models.

Metric	P	B	S_{bad}	S_1	S_2
Accuracy	0.8487	0.8862	0.8833	0.8784	0.8804
Precision	0.8558	0.8865	0.8850	0.8784	0.8816
Recall	0.8487	0.8862	0.8833	0.8784	0.8804
F1 score	0.8479	0.8862	0.8837	0.8784	0.8807

Table XV reports the per-class breakdown. Across all five models, *Business* is consistently the weakest class and *Sports* the strongest; the teacher in particular struggles with *Business* (recall 0.7047), which it frequently confuses with *Science* (344 of 1900 Business articles predicted *Science*, out of 562 total teacher errors on that class). All four students reduce this confusion relative to the teacher (e.g. B: 224 Business→Science errors; S_1: 222; S_2: 227; S_{bad}: 206), consistent with their higher overall accuracy, though none eliminates it.

Table XV: Per-class Precision (P), Recall (R), and F1 for all five AG News models.

Class		P	B	S_{bad}	S_1	S_2
World	P	0.776	0.901	0.929	0.892	0.921
	R	0.873	0.891	0.860	0.893	0.873
	F1	0.821	0.896	0.893	0.892	0.896
Sports	P	0.974	0.948	0.946	0.946	0.951
	R	0.878	0.949	0.948	0.946	0.951
	F1	0.924	0.948	0.947	0.946	0.951
Business	P	0.859	0.859	0.834	0.832	0.823
	R	0.705	0.831	0.848	0.826	0.833
	F1	0.774	0.845	0.841	0.829	0.828
Science	P	0.815	0.839	0.830	0.844	0.832
	R	0.939	0.874	0.876	0.848	0.866
	F1	0.872	0.856	0.853	0.846	0.849

Table XVI reports average KL divergence to the teacher, computed over the full **test** set. Unlike accuracy, KL divergence follows the expected monotonic ordering with distillation strength, $S_2 < S_1 < S_{bad} < B$, at both temperatures – the class-restricted student S_2 is the most faithful to the teacher's output distribution, and the non-distilled B the least, by a wide margin. This is the clearest evidence in this study that, whatever its effect on accuracy, distillation reliably and monotonically pulls the student's output distribution toward the teacher's.

Table XVI: Average KL divergence to the teacher P, computed over the full **test** set (7600 examples), at temperatures $T = 1$ and $T = 2$.

	B	S_{bad}	S_1	S_2
$KL_{T=1}$	2.0802	0.6298	0.2786	0.2477
$KL_{T=2}$	1.7844	0.4183	0.2285	0.2465

2) Functional Evaluation and Explanation Comparison: Figure 9 illustrates **LIME** explanations for a representative *Business*-class example, across all five models – *Business* is chosen as the illustrative class given its weak performance above. As on IMDb, the teacher's explanation is qualitatively more topic-relevant than any student's, and this qualitative impression is again confirmed systematically below.

Tables XVII, XVIII, and XIX report the aggregate agreement metrics over $N = 100$ held-out examples ($k = 10$), analogous to Section IV-A3. Here, the teacher's own agreement-with-itself is trivially 1 (or undefined for faithfulness-independent metrics) and is omitted, reporting only its faithfulness.

Table XVII: **LIME** aggregate agreement metrics (vs. teacher), AG News, $N = 100$, $k = 10$.

Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.138
B	0.336	0.321	0.418	0.758	0.306
S_{bad}	0.348	0.280	0.405	0.762	0.402
S_1	0.342	0.307	0.436	0.791	0.299
S_2	0.358	0.343	0.457	0.814	0.432

Table XVIII: **SHAP** aggregate agreement metrics (vs. teacher), AG News, $N = 100$, $k = 10$.

Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.003
B	0.315	0.270	0.337	0.723	−0.003
S_{bad}	0.317	0.275	0.344	0.736	0.000
S_1	0.320	0.267	0.321	0.722	0.000
S_2	0.311	0.262	0.344	0.715	−0.001

Table XIX: **Integrated Gradients** (zero baseline) aggregate agreement metrics (vs. teacher), AG News, $N = 100$, $k = 10$.

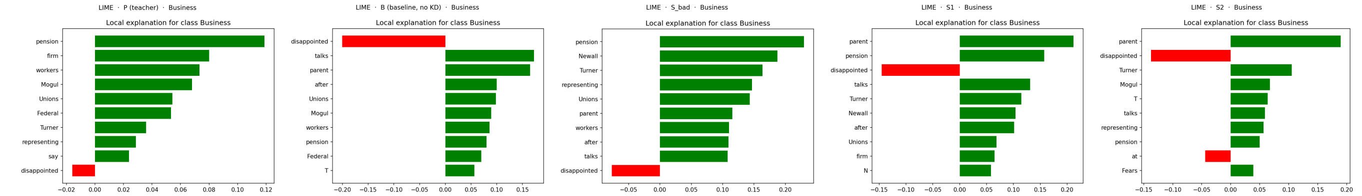
Model	Jaccard@10	Spearman	Pearson	Sign agr.	Faithf.
P	–	–	–	–	0.054
B	0.891	0.022	−0.153	0.509	0.067
S_{bad}	0.883	0.040	−0.087	0.528	0.163
S_1	0.876	−0.036	0.004	0.470	0.138
S_2	0.888	0.013	−0.247	0.504	0.340

The pattern observed on IMDb replicates on AG News: S_2 is the best- or near-best-aligned student on **LIME** and **SHAP** across nearly every agreement metric, and clearly the most faithful of the four on both **LIME** and **IG**; B and S_1 are again the weakest. **SHAP** faithfulness is close to zero (even slightly negative) for all four students and for the teacher itself, indicating that masking the top-10 SHAP-attributed words barely changes the predicted class probability for any model on this task – a pattern discussed further as a possible limitation of the faithfulness metric itself in Section IV-D. As on IMDb, **IG** again shows high top-k overlap (0.876–0.891) alongside near-zero or negative rank/value correlations, reinforcing that IG's apparent agreement across models is driven by selecting similar token sets rather than similar importance orderings.

Table XX reports **LIME** stability. The teacher is again the most stable, and S_1 the least, mirroring IMDb.

3) Representational Comparison:

a) Layer-wise Comparisons: Figure 10 and Table XXI report the layer-wise cosine similarity between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i + 1$).



(a) P (b) B (c) S_{bad} (d) S_1 (e) S_2

Figure 9: **LIME** explanations for a representative *Business*-class AG News example, across all five models, computed using 5000 random word-removal perturbations of the input text.

Table XX: **LIME** stability (mean pairwise Jaccard@10 across 5 seeds, $N = 20$), AG News.

Model	P	B	S_{bad}	S_1	S_2
Stability	0.774	0.669	0.639	0.615	0.688

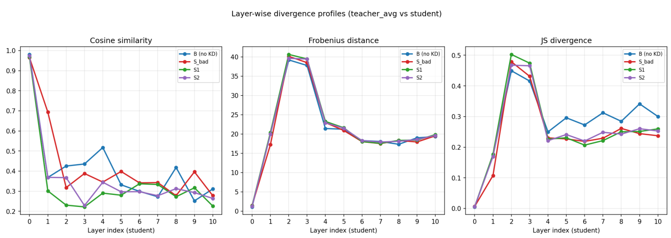


Figure 10: Layer-wise cosine similarity, Frobenius distance, and JS divergence between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i + 1$), for all four AG News students.

As on IMDb, all four models start near-identically similar to the teacher at $i = 0$ (≥ 0.96). From $i = 1$ onward, however, the pattern found on IMDb does *not* replicate: B is *not* consistently the least similar to the teacher. At $i = 1$, S_{bad} is the most similar (0.694) and B, S_1, S_2 are all similarly low (0.30–0.37); at several deeper layers (e.g. $i = 4$: $B = 0.517$, the highest of the four; $i = 8$: $B = 0.418$, again the highest), the non-distilled baseline is at least as similar to the teacher's attention as any distilled student. This is the sharpest finding of this work regarding representational transfer: on the more complex AG News task, distillation does not reliably confer greater attention-level similarity to the teacher than a non-distilled model of the same reduced architecture achieves on its own.

The Frobenius distance (Table XXII) and row-averaged JS divergence (not tabulated, following the same layer-wise profile as Figure 10) tell the same story as on IMDb: all four models are nearly indistinguishable from one another on these metrics, with the familiar profile of near-perfect alignment at $i = 0$ (≤ 1.5, $\ll 2\%$ of the bound of 64), a peak around $i = 2$–3 (≈ 37–41, ≈ 58–63% of the bound), and stabilization around 17–20 in later layers.

Spectrally (not tabulated in full for brevity), the teacher's P_{avg} is again close to rank-one throughout, P_r is highly dispersed ($ER \approx 1650$–2026), and all four students occupy an intermediate, layer-increasing position, with no consistent ordering among B, S_{bad}, S_1, S_2 – the same qualitative picture as on IMDb.

Table XXI: Cosine similarity between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i + 1$), AG News.

i	B	S_{bad}	S_1	S_2
0	0.9809	0.9648	0.9679	0.9748
1	0.3683	0.6935	0.3014	0.3689
2	0.4252	0.3180	0.2305	0.3674
3	0.4354	0.3881	0.2217	0.2284
4	0.5175	0.3454	0.2904	0.3445
5	0.3318	0.3981	0.2800	0.2970
6	0.2992	0.3415	0.3367	0.2977
7	0.2722	0.3428	0.3333	0.2775
8	0.4182	0.2777	0.2723	0.3127
9	0.2517	0.3962	0.3168	0.2929
10	0.3111	0.2778	0.2262	0.2640

Table XXII: Frobenius distance between each student's attention map i and the teacher's P_{avg} (layers $2i, 2i+1$), AG News.

i	B	S_{bad}	S_1	S_2
0	1.08	1.46	1.39	1.22
1	19.94	17.22	20.39	19.95
2	39.22	40.16	40.64	39.64
3	37.78	38.51	39.48	39.45
4	21.42	22.96	23.26	22.91
5	21.24	20.92	21.62	21.50
6	18.28	18.01	18.03	18.27
7	18.05	17.52	17.57	17.91
8	17.33	18.31	18.34	18.10
9	19.00	17.93	18.42	18.58
10	19.33	19.52	19.82	19.62

Finally, cross-student comparisons (not tabulated) show that S_{bad}, S_1, and S_2 remain considerably more similar to *each other* (cosine 0.87–0.98 across layers) than any of them is to the teacher (as low as 0.22–0.40 in mid layers), while B is comparably similar to the three distilled students (cosine 0.66–0.98) as they are to each other – further evidence that the three distillation variants converge toward a shared representational regime that is not obviously closer to the teacher than B's independently-trained one.

b) Global Comparisons: Table XXIII reports global comparisons. As on IMDb, the students' rollout representations align strongly with the teacher's global average ($\cos \geq 0.97$), while their average maps align only moderately ($\cos \approx 0.33$–0.44); as at the layer-wise level, B participates in this pattern to a similar degree as the distilled students; the cosine

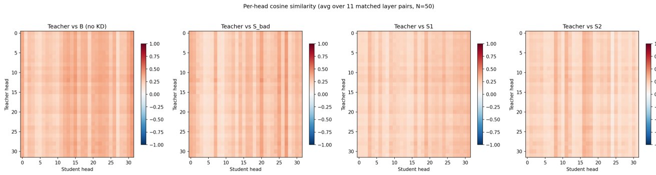


Figure 11: Per-head cosine similarity between teacher and student attention heads, averaged over 11 matched layer pairs and $N = 50$ prompts, for all four AG News students.

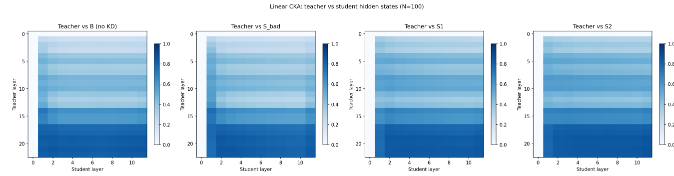


Figure 12: Linear CKA between teacher and student hidden states (all layers, including the embedding layer, $N = 100$ prompts), for all four AG News students.

similarity for $S_{1,avg}$ vs. P_{avg} could not be recovered due to the same type of logging error found in the IMDb notebook.

Table XXIII: Global attention comparisons (whole-model average and rollout maps) against the teacher's global average (P_{avg}) and rollout (P_r), AG News.

Comparison	Cosine	d_F
B_{avg}–P_{avg}	0.4085	20.63
B_{avg}–P_r	0.3204	43.69
B_r–P_{avg}	0.9856	11.34
B_r–P_r	0.9907	13.59
$S_{bad,avg}$–P_{avg}	0.3995	20.76
$S_{bad,avg}$–P_r	0.2981	43.92
$S_{bad,r}$–P_{avg}	0.9811	10.07
$S_{bad,r}$–P_r	0.9854	15.61
$S_{1,avg}$–P_{avg}	–	21.34
$S_{1,avg}$–P_r	0.1898	44.53
$S_{1,r}$–P_{avg}	0.9786	7.40
$S_{1,r}$–P_r	0.9804	18.94
$S_{2,avg}$–P_{avg}	0.3271	21.18
$S_{2,avg}$–P_r	0.2231	44.33
$S_{2,r}$–P_{avg}	0.9745	5.66
$S_{2,r}$–P_r	0.9741	22.18

Spectrally at the global level (not tabulated in full), the teacher's global rollout is close to rank-one, while every student's average map is markedly more dispersed (ER in the low hundreds), again with B comparable to the distilled students rather than clearly lower.

4) Per-Head Attention Similarity and Hidden-State Comparison: Figure 11 and Figure 12 report the per-head cosine-similarity and linear-CKA analyses (Section III-B6), computed identically to their IMDb counterparts (Section IV-A5, Section IV-A6). Both replicate the IMDb findings closely: per-head similarity is diffuse and modest ($\approx$ 0.2–0.5) with a vertical-band structure, and, unlike on IMDb, B's per-head heatmap is not visibly weaker than the distilled students' – consistent with the layer-wise finding above that distillation does not clearly improve attention-level alignment on this task. The CKA heatmaps again show the same near-universal, depth-dependent saturation pattern (low CKA with early teacher layers, near-1 CKA with deep teacher layers regardless of student layer) for all four students, B included, essentially indistinguishable from one another – reinforcing, on a second dataset, that hidden-state geometric similarity to the teacher is largely insensitive to whether a model was distilled at all.

C. Does Distillation Add Value Beyond Depth Reduction?

The model family of Table I, evaluated identically on both datasets, allows us to directly address the question motivating the baseline models introduced in this revision: how much of the behavior previously attributed to distillation is actually attributable to training a smaller model on the same task, independently of distillation? The answer differs by which property is examined.

Classification accuracy is task-dependent. On IMDb, distillation helps substantially: S_2 far outperforms the non-distilled baseline (0.9343 vs. 0.8575 accuracy, Table II). On AG News, the opposite holds: the non-distilled baseline outperforms every distilled variant, including S_2 (0.8862 vs. 0.8804, Table XIV). Whether distillation improves task accuracy beyond depth reduction alone cannot, therefore, be assumed to generalize across tasks.

Distributional fidelity (KL **divergence to the teacher)** is the one property that behaves consistently and predictably: on both IMDb ($KL_{T=2}$: $B = 1.69$ vs. $S_1, S_2 \approx 0.13$–0.19, Table III) and AG News ($KL_{T=2}$: $B = 1.78$ vs. $S_1, S_2 \approx 0.23$–0.25, Table XVI), the non-distilled baseline is by far the least faithful to the teacher's output distribution, and fidelity increases monotonically with distillation strength ($B > S_{bad} > S_1 \gtrsim S_2$ or $B > S_{bad} > S_1 > S_2$). This is distillation's most robust, reproducible effect across both tasks, and it holds regardless of whether accuracy improves.

Explanatory alignment is real but modest. On both datasets, S_2 is consistently the best- or near-best-aligned student with the teacher on **LIME** and **SHAP** agreement and faithfulness metrics (Tables IV–V and XVII–XVIII), and B is typically among the weakest. However, the gap between B and the best-aligned student is small relative to the KL gap above, and **Integrated Gradients** shows no consistent ordering by distillation strength at all on either dataset.

Representational (attention and hidden-state) alignment shows the weakest and least consistent evidence for a distillation-specific effect. At the level of full attention maps, B is clearly the least similar to the teacher on IMDb (Section IV-A4) but is comparable to, or in several layers more similar than, the distilled students on AG News (Section IV-B3). At the level of individual attention heads and of hidden-state geometry (linear CKA), B is visually indistinguishable from the distilled students on *both* datasets (Sections IV-A5–IV-A6 and their AG News counterparts) – the clearest indication in this work that much of the previously reported "teacher-student representational similarity" reflects shared architecture, embeddings, and task exposure rather than a specific product of the distillation objective.

Taken together, these results indicate that knowledge distillation's one dependable contribution, across tasks and independently of whether it improves accuracy, is to pull the student's output distribution toward the teacher's; its effects on classification accuracy, explanatory alignment, and internal representational similarity are comparatively small, inconsistent across datasets, and in the case of fine-grained representational similarity, largely absent once a non-distilled baseline of the same architecture is available for comparison.

D. Limitations

Earlier iterations of this work lacked a model sharing the students' architecture but trained without distillation; this revision addresses that gap directly with B and S_{bad} (Section IV-C). Even with these controls in hand, however, several limitations remain. Only two points along the distillation-strength spectrum were tested beyond the two original students ($\alpha = 1$ for B, $\alpha = 0.9$ for S_{bad}); a denser sweep over α, β, and the temperature T would be needed to characterize the transition between these regimes more precisely, and to determine whether the AG News finding that B matches or exceeds the distilled students' attention-level similarity to the teacher (Section IV-B3) holds at other points in this space. Additionally, B and S_{bad} are trained on the same data and for the same number of epochs as S_1 and S_2, but the four variants are not otherwise matched for compute or number of gradient updates, since the distillation terms add computational overhead per step; a compute-matched comparison was outside the scope of this work.

A further limitation is the hardware available for this work: all training and evaluation was carried out on a single consumer GPU (an NVIDIA RTX 5070 Ti with 16GB of VRAM), which was the binding constraint on the scope of the study – it determined how many datasets, how large a scale for each dataset, and how many additional model or teacher-architecture variants could be explored within a reasonable timeframe, more so than any of the methodological choices discussed above. Relatedly, all experiments use a single teacher architecture (TinyLlama-1.1B); while nothing in our results suggests that this choice is as binding a constraint as compute, evaluating additional teacher architectures, and additional or larger datasets, remains a natural direction for future work once more compute is available.

The explanation-agreement and faithfulness metrics introduced in this revision (Section III-B7) raise a conceptual question also noted by an early reviewer of this work: when **LIME/SHAP** explanations disagree between teacher and student, or when a faithfulness or rank-correlation metric returns a low or inconsistent value, it is not always possible to attribute this cleanly to a failure of distillation as opposed to a limitation of the explanation method itself. Two observations in this study point in this direction. First, **SHAP** faithfulness is close to zero (and occasionally negative) for *every* model, including the teacher, on both datasets (Tables V and XVIII) – since this pattern is not specific to the distilled students, it may reflect a mismatch between the masking-based perturbation used to compute faithfulness and the behavior of a decoder-only generative-classification head, rather than a genuine absence of faithful explanations. Second, **Integrated Gradients** consistently shows high top-k token overlap alongside near-zero rank and value correlation across every model and both datasets, a combination that is more naturally read as a property of the (zero-baseline) IG method on this architecture than as a finding about distillation specifically. Disentangling genuine distillation effects from such method-specific artifacts would require validating each explanation method against a known ground truth, which falls outside the scope of this work.

The per-head attention-similarity and hidden-state (linear CKA) comparisons (Sections IV-A5–IV-A6 and their AG News counterparts) are presented qualitatively, as heatmaps, since the underlying per-head and per-layer-pair numeric matrices were not retained in exportable form during the corresponding experiments; a fully quantitative version of these comparisons (e.g. reporting the specific head pairs or layer pairs that drive the observed patterns) is left for future work.

Finally, the qualitative explanation figures (Figures 5 and 9) illustrate one representative example per dataset rather than the full test set; this is now a much smaller concern than in earlier iterations of this work, since the aggregate metrics of Section III-B7 independently quantify explanatory consistency over $N = 100$ held-out examples per dataset, but the qualitative figures themselves should still be read as illustrative rather than representative in a statistical sense.

V. Conclusions

This work investigated whether the functional similarity induced by knowledge distillation extends to internal representations and explanatory mechanisms in autoregressive Transformers, using a family of models – a non-distilled baseline, a weakly-distilled variant, and two distilled students – evaluated on an initial IMDb pilot and, as the main study, on the more complex, non-binary AG News task.

Regarding explanations, the answer is clearly negative on both datasets: every student, including the non-distilled baseline, assigns substantial importance to semantically or topically irrelevant tokens, and the systematic aggregate metrics introduced in this revision confirm that this is not an artifact of the particular examples chosen. Strong task accuracy does not imply meaningful explanations, nor does it consistently imply better explanatory agreement with the teacher: S_2 tends to be the best-aligned and most faithful student on **LIME** and **SHAP**, but the gap to the baseline is modest, and **Integrated Gradients** shows no consistent ordering at all.

Regarding internal representations, the addition of a non-distilled baseline substantially weakens the case for a distillation-specific effect that earlier iterations of this work could only speculate about. First-layer attention maps remain nearly identical across every model, distilled or not, on both datasets – a property of the shared architecture and embeddings rather than of distillation. At the level of individual attention heads and of hidden-state geometry (linear CKA), the non-distilled baseline is visually indistinguishable from the distilled students on *both* datasets. Only at the level of full, layer-averaged attention maps does a distillation-specific effect appear, and even then inconsistently: distillation helps on IMDb, where the baseline is the least similar model to the teacher throughout, but not on AG News, where the baseline matches or exceeds the distilled students' similarity to the teacher in several layers.

The one property that behaves consistently and predictably across both datasets, independently of whether it improves accuracy, is distributional fidelity: every distilled student is far closer to the teacher's output distribution (KL divergence) than the non-distilled baseline, and fidelity increases monotonically with distillation strength. Taken together, these results support a more precise version of this work's original question: functional similarity in the sense of matching the teacher's output distribution *is* reliably induced by distillation, but classification performance, explanatory alignment, and fine-grained representational alignment are largely dissociable from it and from each other, and, as discussed in Section IV-D, some of the residual agreement or disagreement observed for specific explanation methods may reflect properties of those methods rather than of distillation itself. Future work should extend the (α, β, T) sweep begun here, quantify the per-head and hidden-state comparisons numerically rather than visually, and test whether these conclusions generalize to further tasks and architectures.

REFERENCES

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4190–4197, 2020.

[2] Anthropic. Claude sonnet 4.6 [Large language model]. https://www.anthropic.com, 2025. Used as a generative AI tool for language editing and writing assistance.

[3] Shenghong Dai, Shubham Mehrotra, Shiva Kumar Pentyala, Yingchi Liu, Suman Banerjee, James Zhu, Bin Bi, and Sitaram Asur. Llm as a classifier: Leveraging large language models for text and vision classification. arXiv preprint arXiv:2311.17782, 2024.

[4] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. ERASER: A benchmark to evaluate rationalized NLP models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4443–4458, 2020.

[5] Jianping Gou, Baosheng Yu, Stephen J Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International journal of computer vision*, 129(6):1789–1819, 2021.

[6] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[7] Sarthak Jain and Byron C Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556, 2019.

[8] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529. PMLR, 2019.

[9] Jianhua Lin. Divergence measures based on the shannon entropy. *IEEE Transactions on Information theory*, 37(1):145–151, 1991.

[10] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.

[11] Andrew L. Maas, Raymond E. Daly, Peter T. Pham, Dan Huang, Andrew Y. Ng, and Christopher Potts. Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 142–150, Portland, Oregon, USA, June 2011. Association for Computational Linguistics.

[12] OpenAI. Chatgpt. https://chat.openai.com, 2026. Used as a generative AI tool for language editing and writing assistance.

[13] Stefano Recanatesi, Serena Bradde, Vijay Balasubramanian, Nicholas A Steinmetz, and Eric Shea-Brown. A scale-dependent measure of system dimensionality. *Patterns*, 3(8), 2022.

[14] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should i trust you?" explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*, pages 1135–1144, 2016.

[15] Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. In *2007 15th European signal processing conference*, pages 606–610. IEEE, 2007.

[16] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In *International conference on machine learning*, pages 3319–3328. PMLR, 2017.

[17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[18] Sarah Wiegreffe and Yuval Pinter. Attention is not not explanation. In *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pages 11–20, 2019.

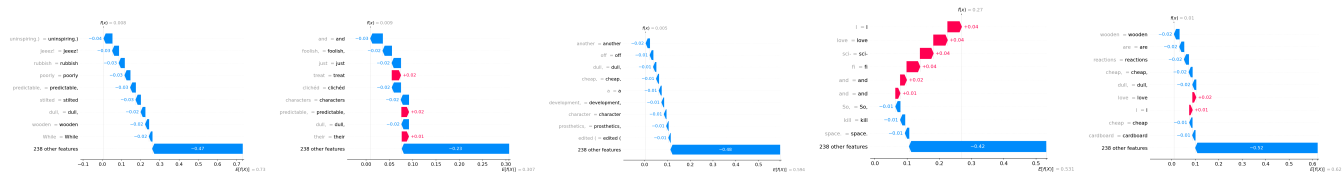
[19] Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. Tinyllama: An open-source small language model. *arXiv preprint arXiv:2401.02385*, 2024.

[20] Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*, volume 28, 2015.

VI. Appendix

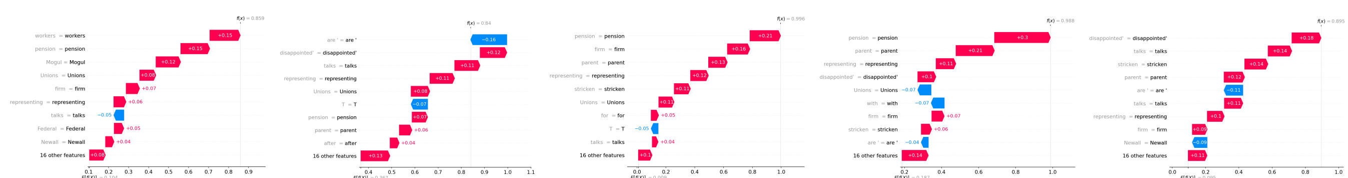
A. ***SHAP*** *Results*

Given that the aggregate metrics of Section III-B7 already quantify **SHAP** and **Integrated Gradients** consistency systematically, this appendix retains only illustrative **SHAP** figures for the same representative examples used in Figures 5 and 9, for all five models.



(a) P (b) B (c) S_{bad} (d) S_1 (e) S_2

Figure 13: **SHAP** explanations for the same representative negative-sentiment IMDb example as Figure 5, across all five models.



(a) P (b) B (c) S_{bad} (d) S_1 (e) S_2

Figure 14: **SHAP** explanations for the same representative *Business*-class AG News example as Figure 9, across all five models.